

Lab 2 — Sprint Plan, Estimation & Reflection

Restaurant Table Booking & Pre-Ordering App | Problem Statement #36

PES University — Dept. of CSE • Name: _____ SRN: _____ Date: _____

1. Sprint 1

Sprint goal: A diner can find a suitable table on the live floor plan, hold it, and browse the menu — with double booking made impossible.

Duration: 1 week • Committed: 34 story points • Completed: 29 • Carried over: 5

ID	User story	Epic	Priority	SP	End-of-sprint status
1.1	View live floor plan	Epic 1	High	8	Done
1.2	Filter tables by slot and party size	Epic 1	High	5	Not completed → carried over
1.3	Select and hold a table	Epic 1	High	5	Done
1.4	Reject overlapping reservations	Epic 1	Highest	8	Done
2.1	Browse the digital menu	Epic 2	High	5	Done
2.4	Show pre-order on the manager terminal	Epic 2	High	3	Done

2. Sprint 2

Sprint goal: A diner can complete a paid pre-order end to end, and the system knows when the kitchen must start cooking.

Duration: 1 week • Committed: 37 story points (32 planned + 5 carried over) • Completed: 37

ID	User story	Epic	Priority	SP	End-of-sprint status
1.2	Filter tables by slot and party size (carried over)	Epic 1	High	5	Done
2.2	Add items to the pre-order cart	Epic 2	High	5	Done
2.3	Edit or remove cart items	Epic 2	Medium	3	Done
3.2	Compute cook-start times	Epic 3	High	8	Done
4.1	Pay the pre-order through the gateway	Epic 4	Highest	8	Done
4.2	Store payment tokens, never card data	Epic 4	Highest	5	Done
4.3	Receive a digital confirmation with QR code	Epic 4	High	3	Done

3. Velocity

Sprint	Committed	Completed	Achieved	Outcome
Sprint 1	34	29	85 %	5 pts (Story 1.2) carried into Sprint 2
Sprint 2	37	37	100 %	Nothing carried over; backlog down to 33 pts
Average	35.5	33	—	Working velocity ≈ 33 pts per 1-week sprint

4. Planning Poker record

Each story was discussed briefly, three estimators voted at the same time, and outliers explained their reasoning before a consensus number was agreed. The five stories below produced the widest spread.

Story	Votes	Agreed	What the discussion surfaced
1.1 View live floor plan	8 / 5 / 13	8	Rendering a live, colour-coded plan is more than a static image — the 13 vote flagged real-time refresh, which the team agreed sits in story 5.2.
1.4 Reject overlapping reservations	5 / 8 / 8	8	Concurrency control is the highest-risk item in the sprint; the team took the higher estimate because of uncertainty, not volume of code.
2.4 Show pre-order on manager terminal	3 / 3 / 5	3	Read-only view over data that already exists — quick agreement.
3.2 Compute cook-start times	5 / 8 / 13	8	Spread came from edge cases (items sharing a station, overlapping bookings); parked as a spike rather than inflating the estimate to 13.
4.2 Store payment tokens, never card data	3 / 5 / 8	5	Little code, but compliance verification adds effort — the classic case for estimating uncertainty rather than hours.

5. Burndown analysis

See [Lab2_Burndown_Charts.pdf](#) for both sprints. Reading the Sprint 1 chart: the grey guideline is where the team should be assuming linear progress; the red line is the work actually remaining. The red line sits above the guideline for the whole sprint, so the team was behind its forecast throughout — not by a lot, but consistently, and it finished 5 points short. Sprint 2 tracks the guideline closely and reaches zero on the last day.

6. Reflection

1. Did your estimations reflect the actual effort?

Broadly yes, but not evenly. The two 8-point stories that were sized for risk rather than volume — 1.4 Reject overlapping reservations and 3.2 Compute cook-start times — landed almost exactly on their estimates, which suggests the team was right to price uncertainty rather than lines of code. The miss was Story 1.2 Filter tables by slot and party size, estimated at 5 and carried into Sprint 2: the estimate covered writing the filter, not wiring it to the live availability service. Sprint 2 estimates held much better because they were re-based on what Sprint 1 actually delivered instead of on optimism.

2. Was your backlog well-prioritized?

The value ordering was sound — every Highest item (1.4 double-booking prevention, 4.1 payment, 4.2 tokenisation) was delivered inside the first two sprints, and the Low item (4.4 promo code) never competed with them. The weakness was dependency ordering rather than value ordering: 5.2 Sync status changes in under 500 ms sits in the backlog even though 1.1 View live floor plan quietly depends on it, so the floor plan shipped against polled data. Priority answered "what matters most" but not "what must exist first".

3. How did your simulated sprint align with your plan?

Sprint 1 committed 34 points and completed 29 (85 %). The burndown stayed flat through Day 1 and Day 2 because all six stories were started in parallel — work was in progress but nothing was accepted, so nothing burned. The line then dropped in large steps as whole stories were accepted, and finished above the guideline with 5 points unfinished. Sprint 2 committed 37 points including the carry-over and finished on Day 7, tracking the guideline far more closely because the team limited work in progress and pulled the next story only when one was accepted.

4. What insights did the burndown chart give about your team's capacity?

Averaged over the two sprints the team burns about 33 points per one-week sprint, so the 33 points still in the backlog are one more sprint of work — a forecast we could not have made before Sprint 1. The step shape is the second insight: points arrive in 5- and 8-point jumps late in the sprint, which is a symptom of stories that are too large to give early feedback. Splitting the remaining 8-point items (5.2) into smaller slices would make the line smoother and any slippage visible on Day 3 instead of Day 6. The chart never fell below the guideline early, so the team is not under-committing.